

Traceability Matrix

Requirement	Related Use Case	Fulfilled By	Test	Description
The application interface contains buttons and display	UC1, UC2, UC6	mainwindow.ui device.ui	Run the simulator in Qt to observe the ui	mainwindow.ui contains buttons for user actions and a box representing the device. device.ui contains buttons on the device and a main screen
The main screen can display a menu	UC1, UC2, UC6	menuscreen.ui	Turn on the device and press the menu button	When a menu is to be displayed, device display the menu screen with a list of items to select
The main screen can display session logs and a date and time setting	UC1, UC6	logdisplayscreen.ui timeinputscreen.ui	Select the corresponding items in the menu	Device choose which screen to display based on the selected item in menu
Users can navigate the system	UC1, UC2, UC6	Device	Press the arrow buttons and the select button on the device	Arrow buttons and select button allows user to navigate the system. Device decides how each button works depending on the displaying screen.
Pressing the new session option opens a timer that begins once contact is initiated	UC2	Neureset Sessions	Select new session in the menu and press the connect button	After the new session option is selected, neureset checks if the connection is established and creates a new session in Sessions.

Initiation of contact is indicated by a blue light	N/A	Device Neureset	Select new session in the menu and press the connect button	If the headset is connected and a new session is created, neureset tells Device to turn on the blue light.
If contact is lost, the red light flashes. The session is paused and the devices starts beeping until contact is reestablished	UC2	Device Neureset	After a session has started, press the disconnect button	When the headset is disconnected, Neureset pauses the session and tells Device to flash a red light and starts beeping until the connection is reestablished or the session is terminated.
If contact is not reestablished in 5 minutes, the session is terminated and device turns off	UC2	Sessions	After a session has started, press the disconnect button or pause button and wait for 5 minutes	When a session is paused whether by disconnection or by user, a timer of 5 minutes is started. The timer can be stopped by resuming or connecting. When the timer runs out, it stops the current session and turns off the device. No data will be logged from that session.
The approx. time remaining and session progress bar	UC4	Device SessionScreen	Start a session with the device	When a session has started, Device will switch screen to session screen, which contains a progress bar and a time display. The progress bar is updated periodically by getting elapsed time from Sessions. Time display gets an estimated time from Sessions from the start and updates with it's own timer.

The user can pause voluntarily during a session, and the session is terminated after 5 minutes	N/A	Device Neureset Sessions	Press the pause button after the session has started	When the pause button is pressed, Device informs Neureset about the event, and Neureset informs Sessions to pause the session and start a timeout timer. Sessions will stop itself and turns off the device after the timeout timer has ran out. Resuming the session will stop the timeout timer.
At the beginning of a session, overall baseline is calculated for all EEG sites	UC3	Sessions EEGSensor	Start the session	When a new session is created, Sessions will start a one minute timer. Once the timer is over, it will collect the baselines of each EEG site and calculates the average baseline of them. The baseline of a site is calculated by EEGSensor using amplitudes and frequencies.
A baseline is calculated for each EEG site individually for 1 minute, then a treatment is applied for a second	UC3, UC4	Sessions EEGSensor	Start the session and wait until the pre baseline collection is over	After the pre baseline is collected and stored, Sessions will loop through all sites, one at a time, collecting individual baseline and applying treatment to them. The treatment apply is performed by EEGSensor, where it adds 5 Hz to the collected baseline and output it to the frequencies of the waves.
At the end of a session, overall baseline is calculated for all EEG sites	UC5	Sessions EEGSensor	Start the session and wait until all sites are treated	After all sites are treated, Sessions will start collecting an overall overall baseline similar to the pre baseline.

Logs for each session are stored for future access	UC5	Sessions Neureset	Start the session and wait for the session to finish	After post baseline is collected, a log is generated in Sessions and stored in Neureset.
Logs can be displayed on the device with time and date	UC6	Neureset Device LogDisplayScreen	Navigate to the SESSION LOG option in menu and select it	When the SESSION LOG option is selected, Device changes the screen to LogDisplayScreen and LogDisplayScreen fetches for the logs in Neureset. Then it gets all the time and dates from the logs and print it in a list.
PC has a UI	N/A	mainwindow.ui pcscreen.ui	Run the simulator in Qt to observe the ui	mainwindow.ui contains the pc screen pcscreen.ui contains a list for logs and an upload logs button
Logs can be uploaded to a PC	N/A	Neureset PCScreen	Press the Upload Logs button in the PC screen	Full logs will be loaded and displayed in a list when the upload logs button is pressed
User can input the date and time to the device	UC1	Device TimeInputScreen	Navigate to the TIME AND DATE option in the menu and select it	When the TIME AND DATE option is selected, Device changes the screen to TimeInputScreen. The user will be able to set the year, month, day, hour and minute for the time and the device will store it and use it as log time.
User can turn on and off the device	UC1, UC2, UC6	Device Neureset	Press the power button on the device	When the power button is pressed, if the device is on, Session is stopped and all lights are turned off. Then the screen turns black and all buttons on the device are disabled. If the device is off and has battery, the screen turns on.

Battery indicator shows current battery level	N/A	Neureset Device	Turn on the device	When the device is on, Device fetches for the battery level from neureset and changes the battery indicator accordingly.
Battery depletes over time	N/A	Neureset	Turn on the device and wait	Every second, a portion of the battery is removed to simulate battery depletion.
Device turns off when battery is completely depleted	N/A	Neureset Device	Turn on the device and wait	When the battery is completely depleted the device turns off and disables all buttons.